An independently implemented lumped-mass mooring line model with a coupled six-degree-of-freedom body: formulation and validation against four experimental data sets

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Abstract

We describe a C++17 lumped-mass (LM) mooring line solver written from the equations of Paredes (2016), with a seabed contact model, Morison drag and added mass, point elements (clump weights and floaters), a nonlinear/viscoelastic rope law, non-uniform lines, explicit and implicit (error-controlled generalised-α) line integration, a small-angle six-degree-of-freedom (6-DOF) rigid body with explicit or implicit coupling of any number of concurrently solved lines, and linearisation about the static state. The code is verified against closed-form solutions (elastic catenary, vibrating string, point-mass oscillators; second-order convergence) and validated against four experimental data sets: the Chalmers 33 m chain [2], the Paredes wave-energy-converter buoy with three mooring configurations [1], the Azcona et al. submerged chain [3], and the Lopez-Olocco et al. clump-weight chain [4]. For forced-fairlead chain tests the maximum fairlead tension is reproduced to 2.0 % on average over 35 clump-free cases and to 0.8 % and −1.8 % over nine cases each with a clump weight; for the Chalmers data set the regression $r^2 = 0.985$. The study also documents the cases that do *not* agree: a snap-loading case (+29 %, not converged under refinement), a buoy pitch period (−6.2 %), moored-buoy wave responses (heave 10–21 % low, surge over-predicted) and a clump pretension rise that is 3 percentage points below the reported value. No hydrodynamic coefficient was tuned to any comparison.

1 Introduction

Lumped-mass models discretise a mooring line into point masses joined by massless axial springs and dampers. They are the standard tool for line dynamics in floating-platform design, and several implementations exist (for example MoorDyn [5]). Their credibility rests on validation against measurements, which is scarce for small, well-instrumented set-ups where the input data are fully known.

This paper reports a new implementation and a validation campaign built around the principle that every disagreement is reported, no coefficient is tuned after seeing the result, and the quality of the evidence is stated for each comparison (digitised charts, values read by eye from figures, or tabulated data). The contributions are: (i) a complete, documented formulation including the numerical limits that govern the explicit time step; (ii) a verification suite; (iii) four validation comparisons with sensitivity studies; (iv) a candid list of limitations.

The implementation follows the equations of Paredes [1] (equation numbers below refer to that thesis); no source code from any other mooring solver was used.

2 Formulation

2.1 Line discretisation and tension

A line of unstretched length L is divided into N segments of length $l_0 = L/N$. Node i has position $\boldsymbol{r}_i$ and mass $m_l l_0$ (m_l the dry mass per unit length; half-masses at the ends). With segment stretch $\varepsilon_j = |\boldsymbol{r}_{j+1} - \boldsymbol{r}_j|/l_0 - 1$ the tension is bilinear with internal damping (Eq. 3.25),

$$T_j = \max\!\big(0,\; EA\,\varepsilon_j + c_{\mathrm{int}}\dot{\varepsilon}_j\big), \tag{1}$$

so a slack segment carries no load. Stiffness-proportional damping is expressed as $c_{\mathrm{int}} = \beta EA$.

2.2 Environmental loads

The submerged weight per unit length (Eq. 3.26) blends the dry and submerged values about the still-water level. Drag on a node uses the relative water velocity $\boldsymbol{v}_r$ with tangential and normal Morison coefficients over the stretched tributary length (Eqs. 3.28–3.31),

$$\boldsymbol{F}_D = \Big[\tfrac{1}{2}\rho D\, C_{dt}\, v_{rt}|v_{rt}|\,\boldsymbol{t} + \tfrac{1}{2}\rho D\, C_{dn}\,|\boldsymbol{v}_{rn}|\boldsymbol{v}_{rn}\Big]\, l_0(1+\varepsilon). \tag{2}$$

Added mass acts in the direction normal to the line, $m_a = C_m \rho A(1+\varepsilon)l_0$, and the node acceleration is obtained by solving $(m\,\mathbf{I} + m_a(\mathbf{I} - \boldsymbol{t}\boldsymbol{t}^{\mathsf{T}}))\boldsymbol{a} = \boldsymbol{F}$ per node (Eq. 3.27), with the tangent $\boldsymbol{t}$ from the neighbouring nodes. The inverse mass matrix is $\mathbf{M}^{-1} = \boldsymbol{t}\boldsymbol{t}^{\mathsf{T}}/m_t + (\mathbf{I} - \boldsymbol{t}\boldsymbol{t}^{\mathsf{T}})/m_n$.

2.3 Waves and currents

Waves are linear Airy components over constant depth with optional Wheeler stretching. Irregular seas are generated from a JONSWAP or Pierson–Moskowitz spectrum with prescribed H_s and T_p: n equal frequency bins, one component at a (seeded) uniformly random frequency in each bin with amplitude $\sqrt{2S(\omega_i)\Delta\omega}$ and random phase, rescaled so that $\sum a_i^2/2 = H_s^2/16$ exactly. The first-order load on the body is $F_i(t) = \sum_k w_i(\omega_k) a_k \sin(\omega_k t + \phi_k + \delta_i(\omega_k))$ with a frequency table w_i, δ_i interpolated linearly (a component outside the table is rejected); the body added mass and damping remain constant. A steady current, uniform or depth-varying (table or power law), is added to the orbital velocity for the lines and point elements; there is no wave–current interaction and no current load on the body.

2.4 Seabed

Penetration $p > 0$ below the seabed gives a normal spring–damper (Eq. 3.32) and a velocity-ramped Coulomb friction (Eqs. 3.34–3.36),

$$F_n = \big[K_s D_1\, p - 2\zeta\sqrt{K_s D_1 m_l}\,\min(0, v_n)\big] l_0, \qquad \boldsymbol{F}_t = -w_{\mathrm{sub}}\,\mu\,\min\!\big(|\boldsymbol{v}_t|/v_{\mathrm{lim}}, 1\big)\,\hat{\boldsymbol{v}}_t\, l_0, \tag{3}$$

with K_s the soil stiffness per unit length and diameter, ζ the damping factor, μ the friction coefficient and v_{lim} the ramp speed (a numerical parameter).

2.5 Point elements

Clump weights and floaters are attached to a node and contribute weight, buoyancy $\rho g V$, Morison drag (isotropic, or with separate tangential and normal coefficients and areas) and added mass (isotropic, or separate tangential and normal values). The anisotropic option was needed to reproduce the clump model of Lopez-Olocco et al. (normal added mass $\rho D^3/3$).

2.6 Rigid body and coupling

The platform is a 6-DOF rigid body with constant mass-plus-added-mass matrix $\mathbf{A}$, damping $\mathbf{B}$, stiffness $\mathbf{C}$, an optional quadratic drag, a constant load, and small-angle rotational dynamics (Eq. 3.59). Fairlead positions follow from the exact exponential-map kinematics. Any number of lines are attached to the body; at each body step the fairlead forces and moments of all lines are summed (explicit partitioned coupling). The body uses velocity-Verlet with implicit linear damping and the lines are sub-stepped at their own stable time step. Static equilibrium of the body is found by Newton iteration with a finite-difference Jacobian.

Concurrent line solves. The lines of a platform are independent objects between coupling steps, so their static solves (inside the equilibrium iterations), start-up evaluation and sub-stepping of every body step can run on a thread pool. The results are bit-identical to the serial run; the gain is bounded by the number of lines (2.25 for the three-leg Paredes CAT case, 55.8 s to 24.8 s).

Implicit coupling. For stiff lines on light bodies the explicit scheme is limited to $\Delta t_{\mathrm{body}} < 2/\omega_s$, $\omega_s = \sqrt{K_{\mathrm{line}}/M}$. An optional implicit body integration uses Newmark-β for the pose $\boldsymbol{\xi}_{n+1}$ with Newton iterations on $\boldsymbol{R} = \mathbf{M}\boldsymbol{a} + \mathbf{B}\boldsymbol{v} + \mathbf{D}_q|\boldsymbol{v}|\boldsymbol{v} - \boldsymbol{F}_{\mathrm{ext}}(\boldsymbol{\xi}, \mathrm{lines}, t_{n+1})$, the approximate Jacobian $\mathbf{M}/(\beta\Delta t^2) + \gamma\mathbf{B}/(\beta\Delta t) + \mathbf{K}$ with the static linearised stiffness $\mathbf{K}$ (refreshed when convergence is slow), and a re-advance of every line from its saved state at each iteration with the trial fairlead path. It removes the stability limit but costs several line advances per step, so it does not help the validation cases of this paper, which are not stiffness-limited.

2.7 Linearisation

About a static state the code provides (i) the natural modes of a line, $\mathbf{K}\boldsymbol{\phi} = \omega^2\mathbf{M}\boldsymbol{\phi}$ with $\mathbf{K} = -\mathbf{M}\,\partial\mathbf{A}/\partial\boldsymbol{r}$ taken from the finite-difference Jacobian of the implicit solver (symmetrised, dynamic stiffness including the Maxwell branches), nodal inertia including normal added mass, and modal damping ratios $\zeta_k = \boldsymbol{\phi}_k^{\mathsf{T}}\mathbf{C}\boldsymbol{\phi}_k/2\omega_k$; (ii) the 6×6 stiffness of the mooring on the body, $\mathbf{K} = -\partial\boldsymbol{G}/\partial\boldsymbol{\xi}$ by central differences of the static line solutions, the same Jacobian as in the equilibrium iteration; and (iii) the undamped natural frequencies of the body from $(\mathbf{C} + \mathbf{K}_{\mathrm{lines}})\boldsymbol{x} = \omega^2(\mathbf{M} + \mathbf{A})\boldsymbol{x}$. Eigenproblems are solved by the cyclic Jacobi method.

2.8 Rope law

Synthetic ropes are described phenomenologically by a static tension curve $T_s(\varepsilon)$ (piecewise linear through the origin) in parallel with n Maxwell branches, each a spring K_k in series with a dashpot of relaxation time τ_k:

$$T = T_s(\varepsilon) + \sum_k K_k(\varepsilon - \alpha_k) + c_{\mathrm{int}}\dot{\varepsilon}, \qquad \dot{\alpha}_k = (\varepsilon - \alpha_k)/\tau_k. \tag{4}$$

The relaxed stiffness is the curve and the high-rate stiffness adds $\sum_k K_k$, which reproduces a dynamic stiffness above the static one; the complex modulus of a branch is $K\omega^2\tau^2/(1+\omega^2\tau^2) + \mathrm{i}K\omega\tau/(1+\omega^2\tau^2)$. The internal strains α_k are held during a step and advanced exactly for a linearly varying strain. The law has no hysteresis in the static curve, no mean-load dependence of the dynamic stiffness and no creep, and its parameters must come from tests of the rope concerned; *it has not yet been compared with rope measurements.*

2.9 Bending stiffness

Bending is represented by the discrete curvature energy $U_b = \sum_i EI_i|\boldsymbol{t}_i - \boldsymbol{t}_{i-1}|^2/(2\bar{l}_i)$ over the interior nodes, with $\boldsymbol{t}_j$ the unit segment tangents and $\bar{l}_i$ the mean unstretched length of the two

segments, i.e. $\sum \frac{1}{2} EI\kappa^2 \bar{l}$ with $\kappa = \theta/\bar{l}$. The nodal forces are the exact gradient (a five-node stencil, so the Jacobian of the implicit scheme becomes pentadiagonal in blocks and is solved by a banded LU). The ends are pinned by default; a clamped end (built-in end, bend stiffener, hang-off angle) adds $EI(1-(\boldsymbol{c}\cdot\boldsymbol{t})^2)/l_0$ for the end segment. The explicit step gains the limit $\Delta t \le C_{\mathrm{cfl}}\, l_0^2\sqrt{m_l/EI}/2$ from the highest bending frequency $4\sqrt{EI/(m_l l_0^4)}$.

2.10 Non-uniform lines

A line may consist of sections from anchor to fairlead (for example chain–rope–chain), each with its own length, number of segments and properties (stiffness or rope law, mass, weight, diameters, drag and added-mass coefficients, internal damping). Properties are stored per segment and a node takes half of each adjacent segment, so section boundaries need no special treatment. Uniform lines are reproduced bit for bit.

2.11 Time integration and stability

Lines are advanced by classical RK4 (a velocity-Verlet alternative is available). The explicit step is the minimum of three limits: the axial-wave CFL condition $\Delta t \le C_{\mathrm{cfl}}\, l_0/c$, with $c = \sqrt{EA/m_l}$; the internal-damping limit $\Delta t \le C_{\mathrm{cfl}}\, m_l l_0^2/(2c_{\mathrm{int}})$; and the seabed contact oscillator limit based on $\omega^2 = K_s D_1/m_l$. The damping limit is what makes the Lopez-Olocco runs slow ($\Delta t \approx 10^{-5}\,\mathrm{s}$). The initial state is a touchdown catenary relaxed by dynamic relaxation with a fictitious mass.

Two implicit options remove the explicit limits. (i) The internal axial damping can be treated implicitly by a Strang splitting (backward-Euler half steps of the dashpot forces around each explicit step), which lifts the damping limit. (ii) The whole line can be integrated by the generalised-α method (Newmark for $\rho_\infty = 1$), with Newton iterations on the step increment, a finite-difference block-tridiagonal Jacobian exploiting the three-node coupling (reused for several steps), a backtracking line search against the kink of the bilinear tension, and halving of the step on failure; if Newton fails at the smallest step an explicit micro-step is taken, never an unconverged implicit iterate. An optional error control uses the Newmark local error $(\beta - \frac{1}{6})\Delta t^2(\boldsymbol{a}_1 - \boldsymbol{a}_0)$ converted to a segment-strain error and scaled by a tolerance times the peak strain, with step rejection and an elementary controller (factor limits 0.2–2, safety 0.9, minimum step 1/1024 of the maximum).

3 Verification

Table 1 summarises the verification suite (all in the repository test suite; 2274 assertions pass). Cycle maxima taken from short windows carry a round-off noise floor of order 10^{-4} relative (larger for snap events) because slack/contact events amplify round-off; differences below this are not interpreted.

4 Validation

Each comparison states the evidence quality. Acceptance criteria were fixed before the runs where noted.

4.1 Chalmers 33 m chain [2]

A 33 m chain with a circularly driven fairlead (30 cases, radii 0.075–0.2 m, periods 1.25–3.5 s). The measured maxima are digitised chart values with a ±5 % reading error. With 66 segments and the published coefficients, the static top tension is 22.653 N (22.685 N at 132 segments) against the published 22.68 N. The regression of simulated against measured maxima gives $r^2 = 0.985$

(published model: 0.98), RMSE 1.4 N, bias +0.12 N; 24 of 30 cases lie within 5 % and all within 6.7 % (Fig. 1).

Disclosure. The rotation sense is not stated in the inputs. The first full grid was run counter-clockwise (an arbitrary default) and over-predicted by 9.8 % (r^2 against 1:1 = 0.81). The clockwise sense was then adopted after Fig. 3.5 of the thesis was consulted; the check was motivated by the mismatch. The result therefore depends on this inferred set-up detail. Smooth cases converge for $N \geq 33$ (variation ≤ 0.15 %); the snap case ($r = 0.2$ m, $T = 1.25$ s) scatters by ± 1.2 % in N.

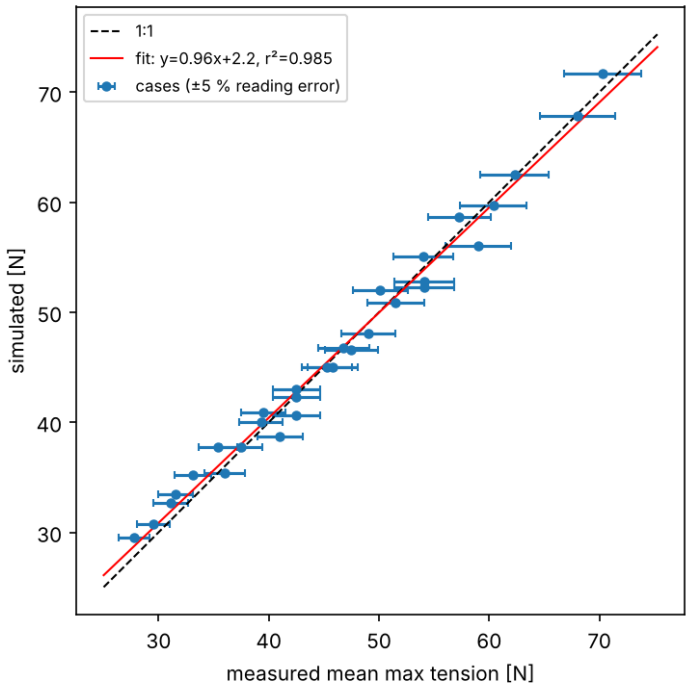


Figure 1: Chalmers grid: simulated against measured maximum tension.

4.2 Paredes buoy [1]

A cylindrical buoy ($m = 35.5$ kg) moored by three legs in three configurations: CON1 (rope with floater), CON2 (with clump weight) and CAT (chain catenary). No coefficient was adjusted; chain hydrodynamic coefficients were applied to the rope (an assumption). Linearised about the static equilibrium, the undamped surge periods are 9.34, 9.16 and 10.02 s for CON1, CON2 and CAT, heave 1.097, 1.095, 1.096 s and pitch 1.074, 1.094, 1.082 s. For CON1 and CON2 they agree with the damped periods of the decay test, as they should (CON2: 9.156 against 9.207 s); for CAT the linear period at rest is 7 % above the decay period because the catenary stiffness depends on amplitude (U-shaped, 33.9 and 20.4 N/m at ∓ 0.1 m). The lowest line modes (periods 0.7–1.5 s) lie close to the wave periods of the RAO tests, which is worth remembering when interpreting the wave-run disagreements below. The statics agree within the measurement error. The pitch period disagrees (-6.2 %; the thesis' own total inertia reproduces 1.170 s but is derived from that test), and the CON1 stiffness is very sensitive to the rope length: a 1.5 % length change moves the stiffness by 13 %, so the disagreement lies within the geometric uncertainty of the as-built model.

Regular waves ($T = 1.3$ and 1.4 s). With the body coefficients of the thesis (Table 3.4) and Wheeler stretching on the lines, heave is 10–21 % low in 6/6 cases (all within the 25 %

criterion fixed in advance), pitch meets the 30 % criterion in 3/6 (CAT 35–50 % low), and surge is over-predicted by 14 % up to a factor three. The measured surge RAO has a sharp minimum near $T = 1.3$ s that neither the moored model nor the free hull reproduces. The wave module itself is verified to 10^{-6}–10^{-12} against closed forms. Submerging the line nodes and point elements at the instantaneous free surface instead of the still-water level changes the responses by at most 1.2 %, so that approximation is not the cause of the disagreements. We regard the wave response as *not validated*.

4.3 Azcona et al. submerged chain [3]

A 21 m chain in a 5 m deep basin with the fairlead driven horizontally ($T = 1.58$, 3.16, 4.74 s). The amplitude in Table 1 of the paper (0.25 m) was interpreted as peak-to-peak (±0.125 m), consistent with the axes of the paper's figures; with 0.25 m as the amplitude the snap case would be 268 N against 44.5 N measured. Static fairlead tension: 8.199 vs. 8.13 N (+0.9 %) and 15.02 vs. 14.48 N (+3.7 %, equivalent to a 2.4 cm anchor shift). Five of six cases meet the ±8 % criterion fixed in advance. With the explicit schemes the snap case is not a converged number: it spans 48–67 N across discretisations and time steps. With the error-controlled implicit integrator (tolerances 10^{-2} to 10^{-5} of the peak strain, 60 segments) the maximum is 49.1, 48.6, 48.5 and 49.4 N, i.e. the time-step scatter at fixed discretisation is about ±1 % and most of the earlier scatter was time-step noise. The remaining dependence on the number of segments is larger, 52.3, 48.5, 51.6 and 50.3 N for 30, 60, 120 and 240 segments, so the converged value is about 50 ± 2 N, $+12 \pm 5$ % above the measurement. The physical inputs the paper does not pin down (anchor position, friction, internal damping) move the result by −10 to −20 %, so the remaining difference is a model or input effect, not the time integrator. Bending stiffness, whose value for a chain is not defined (a solid steel rod of the same diameter would have $EI = 1.4\,\mathrm{N\,m^2}$), was varied as an exploration: it has no effect for $EI \leq 10^{-3}\,\mathrm{N\,m^2}$ and raises the peak above that (58.5, 63.0 and 81.4 N for $EI = 10^{-2}$, 10^{-1} and $1\,\mathrm{N\,m^2}$), so it cannot explain the excess over the measurement.

4.4 Lopez-Olocco et al. clump-weight chain [4]

A 27 m studless chain ($\mathrm{EA} = 3.416 \times 10^5$ N) at 6.5 m depth, anchor distance 25 m, fairlead on a horizontal circle of radius 0.125–0.225 m with period 2.8–5.5 s; a clump (122 g, 80 $\mathrm{cm^3}$) at $L_s/3$ (CW1) or $L_s/2$ (CW2) from the fairlead. Measured maxima and minima are tabulated in the paper (Tables 8 and 9) and were transcribed from its text layer. Soil and internal-damping parameters are the paper's (Table 7; $\beta = 0.0007$ s, zero friction); nothing was tuned. The load-cell accuracy is about ±0.18 N, so the clump-free minima are within instrument accuracy. Static pretension at 25 m is 11.48 N (no clump), 12.17 N (CW1) and 12.27 N (CW2); the paper reports a CW2 rise of about 10 % whereas the model gives +6.9 %, so the static effect of the clump is somewhat under-predicted for a reason not yet isolated. The clump tangential drag coefficient (1.17) is assumed (not in the source); for CW1 at $A = 0.225$ m, $T = 2.8$ s (baseline 24.87 N, measured 24.31 N), $C_{dt} = 0.8$ and 1.5 and no added mass give 24.78, 24.96 and 25.00 N. Only 53 of the 105 cases were run, the clump cases are not converged in segment count, and one tabulated value (A = 0.150 m, CW2, $T = 5.0$ s: 15.72 N) appears to contain a typographical error and was not used.

5 Discussion

What the evidence supports. For taut-to-snapping chains driven at the fairlead, the model reproduces maximum fairlead tension to within a few percent for most cases across three independent experiments, with coefficients taken from the sources. Static properties of a three-leg buoy mooring agree within the stated measurement uncertainty.

What it does not support. (i) Snap loading with total loss of tension is reproduced only to about $+12 \pm 5\,\%$ even when the time integration is converged; the residual is dominated by the spatial discretisation and by inputs the source does not give. (ii) Wave-driven motions of the moored buoy are not reproduced (surge in particular), and the potential causes (hydrodynamic coefficient set, reference point, second-order effects) are untested. (iii) The coverage is limited to chains and one polyester rope at laboratory scale; no full-scale, multi-line platform or synthetic-rope data have been compared. The nonlinear/viscoelastic rope law and the non-uniform line formulation are implemented and verified against closed forms but have not been compared with rope measurements.

Evidence quality. Several reference values are digitised or read by eye (Chalmers chart $\pm 5\,\%$; Azcona figures $\pm 0.5\,\mathrm{N}$; Paredes figures), which limits the discrimination of small model errors; the Chalmers rotation sense was inferred after a mismatch; two interpretation choices (Azcona amplitude, Lopez-Olocco clump drag) were required.

6 Limitations

Cable without torsion or VIV; small-angle rigid-body dynamics with constant $\mathbf{A}, \mathbf{B}$; Euler–Bernoulli bending without torsion, moment–curvature nonlinearity or bending damping; linear waves only (regular or irregular, with no second-order loads, spreading or wave–current interaction, and constant body added mass and damping), with line submergence evaluated at the still-water level by default; flat seabed with a spring–damper; explicit coupling whose body step is limited to about 0.5–0.9 of $2/\omega_s$ (an implicit option exists but is more expensive per step and its accuracy at large body steps is limited by the line's own lowest vibration period); linearisation at a single static state without hydrodynamic damping of the line modes; a single rigid body; piecewise-uniform (not continuously tapered) line properties; a heuristic error estimate for the implicit integrator that is relative to the peak strain.

7 Conclusions and outlook

A newly written lumped-mass solver with a coupled 6-DOF body reproduces fairlead tension in three chain experiments to a few percent without tuning (clump-free chain $+2.0\,\%$ on average over 35 cases, with clump weights $+0.8\,\%$ and $-1.8\,\%$), and the paper documents where it fails. The implicit and error-controlled integration shows that the time step is not responsible for the snap-case disagreement. Planned work: validation against the NREL/NLR 1:100 mooring data and the OC5 data, which would provide multi-line and larger-scale evidence and the first test of the rope law; frequency-dependent body added mass and damping, second-order wave loads and current loads on the body; and torsion and a nonlinear moment–curvature law for dynamic power cables. [TODO: revise once additional experimental data have been compared]

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Check	Result
Static LM line vs. elastic catenary, $N = 10$–160	rel. L_2 error $3.6 \times 10^{-4} \to 1.5 \times 10^{-6}$, order 1.99
Vibrating string, $N = 5$–80 (RK4 and Verlet)	order 2.01
Clump at mid-span, floater vs. exact half-catenaries	order 1.99, error $\leq 6 \times 10^{-5}$
Quadratic drag decay vs. first-order theory	0.1–0.7 % for $t \geq 1$ s
Chain with seabed touchdown vs. analytic H, V	21.387 vs. 21.381 N
Rotation of the set-up by 37°	within the metric noise floor (10^{-4} smooth, ≤ 0.5 % snap)
Rope law: stress relaxation, branch modulus, cable dynamic stiffness	10^{-9}; 2×10^{-3}; E' 0.03 %, E'' 0.5 %
Two-section line: hanging equilibrium; two-medium string frequency	10^{-6}; -0.001 %
Implicit damping: mode decay at 40× the explicit limit	exact rate, monotone dissipation
Generalised-α: vibrating string, error ratio per halved step	4.0 (second order)
Adaptive steps: mode error per decade of tolerance	×10
Concurrent line solves vs. serial (three-leg platform)	identical (bit for bit)
Bending: force against $-\nabla U_b$ (random 3-D shape, clamped ends)	4×10^{-7} N
Bending: pinned / clamped beam deflection; taut beam frequencies 1–3	+0.01 % / +0.49 %; ≤ 0.03 %
Spectra: PM closed form; exact H_s; elevation variance of a 20 000 s record	10^{-4}; 10^{-12}; 0.5618 vs. 0.5625 m^2
Free body in a 30-component JONSWAP sea: heave vs. summed component responses	10^{-4} of the amplitude
Sheared current: Morison drag at every node vs. analytic	exact
Line modes: transverse modes of a taut chain, first longitudinal mode, modal damping	5 digits; 6 digits; $\zeta = \beta\omega/2$
Two-section string, linearised lowest frequency	-0.0007 %
Body on a stiff line: stiffness EA/L_0 and T/L	exact; 0.24 % (line weight)
Implicit coupling: stiff line on a light body at 8× the explicit limit	within 1 % of the small-step reference; explicit diverges
Body on a vertical line (heave) and pendulum	6×10^{-5} and -9×10^{-5} relative error
Three-leg static equilibrium	residual 10^{-9} N; independent balance 10^{-6}
Free buoy in regular waves vs. closed-form RAO	agreement $< 10^{-3}$

Table 1: Verification against closed forms.

Quantity	Model	Measured
Free heave period	1.0997 s	1.112 ± 0.006 s (-1.1 %)
Free pitch period	1.0979 s	1.170 ± 0.005 s ($\mathbf{-6.2}$ %)
Leg tension CON1 / CON2 / CAT	2.75 / 10.9–11.0 / 2.93 N	2.8–3.1 / 10.6–11.0 / 3.0–3.1 N
Draft change CON1 / CON2 / CAT	+0.6 / +12.9 / +3.4 mm	+1 / +13 / +4 mm
Surge stiffness vs. Fig. 5.21	CON2 within 3 %; CAT 7–10 % low; **CON1 15–19 % low**	read by eye, ±1.5 N/m
Damped surge period CON1 / CON2 / CAT	9.286 / 9.207 / 9.349 s	8.561 / 9.22 / 9.14 s

Table 2: Paredes buoy: free decay, statics and surge decay (no tuning).

Config.	T [s]	Model max [N]	Measured [N] (by eye, ±0.5)	Diff.
1	1.58	15.12	14.5	+4.3 %
1	3.16	10.23	9.8	+4.4 %
1	4.74	9.35	9.2	+1.6 %
2	1.58 (snap)	57.37	44.5	**+28.9** %
2	3.16	25.07	23.5	+6.7 %
2	4.74	20.35	19.5	+4.4 %

Table 3: Azcona et al.: maximum fairlead tension (60 segments).

Config.	Cases	Max, mean diff.	Max, worst	Min, mean $\lvert\Delta\rvert$ [N]	Min, worst [N]
No clump	35	+2.04 %	+4.83 %	0.071	0.215
CW1	9	+0.76 %	+2.32 %	0.453	0.911
CW2	9	−1.81 %	−3.70 %	0.174	0.372

Table 4: Lopez-Olocco et al.: maximum and minimum fairlead tension (60 segments without clump, 81 with).